Topological Anti-Gravity: π-Berry Phase Materials as Gravitational Sign Reversers in the Spectral Action Framework

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We show that a material possessing a topological π-Berry phase reverses the sign of the effective gravitational coupling in Connes' spectral action framework. The mechanism is elementary: a π-phase shift applied to all vacuum modes transforms the spectral heat kernel from $\zeta(t)$ to $-\zeta(t)$, reversing the Seeley-DeWitt coefficient a_2 and hence $G_{\text{eff}} = -G$. No prime-by-prime Euler factor manipulation is required; a single global topological phase suffices. Candidate materials include π-Josephson junctions, topological insulator surfaces (Bi_2Se_3), and Weyl semimetals—all experimentally characterized. The anti-gravity effect is *passive*: the Berry phase is geometric, requiring no energy input. We propose two experimental tests: (1) a torsion balance measurement of gravitational anomaly near a π-junction cavity, and (2) the Casimir sign test in a π-junction-bounded resonator. If confirmed, π-Berry phase materials constitute the first known class of "anti-gravity matter"—substances that gravitationally repel rather than attract.

INTRODUCTION

All known matter gravitationally attracts. The weak energy condition (WEC) $T_{\mu\nu}u^\mu u^\nu \geq 0$ holds for all classical matter, and violations require exotic quantum states [**?**]. Here we identify a mechanism that reverses the *coupling* $G \to -G$ rather than the energy density $\rho \to -\rho$, achieving gravitational repulsion with ordinary positive-energy matter.

The mechanism arises in Connes' spectral action principle [**?**], where Newton's constant emerges from the spectrum of the Dirac operator D: $G^{-1} \propto a_2(D^2)$, the second Seeley-DeWitt coefficient of the heat kernel $K(t) = \text{Tr}(e^{-tD^2})$. If the spectrum is globally sign-reversed ($K \to -K$), then $a_2 \to -a_2$ and $G \to -G$.

We show that a material with topological π-Berry phase implements precisely this sign reversal.

THE SIGN-REVERSAL THEOREM

Theorem 1 (Topological anti-gravity)**.** *Let (A, H, D) be a spectral triple with heat kernel $K(t) = \text{Tr}(e^{-tD^2}) = \zeta(t)$ (the Bost-Connes identification [**?**]). If a global π-phase is applied to all modes, the heat kernel becomes*

$$K_\pi(t) = \text{Tr}\big((-1)\, e^{-tD^2}\big) = -\zeta(t), \tag{1}$$

and the effective gravitational coupling reverses:

$$\boxed{G_{\text{eff}} = -G.} \tag{2}$$

Proof. The Seeley-DeWitt expansion for $d = 4$: $K(t) = a_0 t^{-2} + a_2 t^{-1} + a_4 + \cdots$. The gravitational coupling is $G^{-1} \propto a_2 \propto K'(0)$.

For the normal case: $K(0) = \zeta(0) = -\frac{1}{2}$, $K'(0) = \zeta'(0) = -\frac{1}{2}\log(2\pi)$.

For the sign-reversed case: $K_\pi(0) = -\zeta(0) = +\frac{1}{2}$, $K'_\pi(0) = -\zeta'(0) = +\frac{1}{2}\log(2\pi)$.

The ratio:

$$\frac{G_{\text{eff}}}{G} = \frac{K'_\pi(0)}{K'(0)} = \frac{-\zeta'(0)}{\zeta'(0)} = -1. \tag{3}$$

□

The result is exact, not perturbative. Unlike the Liouville rotation [**?**] ($\zeta(2t)/\zeta(t)$, giving $G = -2G$ but requiring infinitely many operations), the global sign reversal requires a *single* operation: multiply every mode by (-1).

PHYSICAL MEANING: BOSONIC → FERMIONIC VACUUM

In quantum field theory, bosonic and fermionic fields contribute with opposite signs to the vacuum energy:

$$E_{\text{bos}} = +\tfrac{\hbar}{2}\sum_n \omega_n = +\tfrac{\hbar\omega_0}{2}\,\zeta(-d), \tag{4}$$

$$E_{\text{fer}} = -\tfrac{\hbar}{2}\sum_n \omega_n = -\tfrac{\hbar\omega_0}{2}\,\zeta(-d). \tag{5}$$

The global π-phase transforms every mode from bosonic (+) to fermionic (−) character. This is not a change of spin or statistics in the particle-physics sense; it is a change of the *vacuum energy sign* at the level of the spectral action.

In supersymmetric theories, the exact cancellation $E_{\text{bos}} + E_{\text{fer}} = 0$ solves the cosmological constant problem. Here we go further: a *fully fermionic* vacuum gives $E_{\text{total}} = -E_{\text{bos}}$, reversing the gravitational coupling.

π-BERRY PHASE MATERIALS

A topological π-Berry phase arises when a wave function acquires a geometric phase of exactly π upon traversal of a closed path in parameter space. This is topologically quantized and robust against perturbations.

Material	Mechanism	Ref.
π-Josephson junction	intrinsic π shift	[?]
Bi_2Se_3 surface	Dirac cone Berry phase	[?]
Graphene (K-point)	sublattice Berry phase	[?]
HgTe quantum well	2D TI edge state	[?]
Weyl semimetal	chiral anomaly	[?]

All candidates are experimentally well-characterized. π-Josephson junctions in particular can be fabricated as thin films [?] and integrated into superconducting circuits.

Implementation

A cavity whose boundary is a π-Berry phase material imposes the condition that every mode acquires a π-phase upon reflection from the boundary. The effective boundary condition is:

$$\psi(x_{\text{boundary}}) = -\psi(x_{\text{boundary}}), \tag{6}$$

which is an *anti-periodic* (fermionic) boundary condition.

The cavity spectral zeta function becomes $\zeta_{\text{cavity}}(t) = -\zeta(t) + O(\text{finite-size corrections})$, implementing the sign reversal of Theorem 1.

The effect is passive: the Berry phase is a property of the material's topology, not of an externally applied field. Once fabricated, the π-junction cavity maintains $G_{\text{eff}} = -G$ without energy input.

EXPERIMENTAL TESTS

Test 1: Casimir sign reversal

In a standard (bosonic-boundary) cavity: $E_{\text{Cas}} < 0$ (attractive Casimir force). In a π-junction cavity: $E_{\text{Cas}} > 0$ (repulsive Casimir force).

Setup. Compare the Casimir force between two plates: (A) both plates normal metal (DD, bosonic), (B) one or both plates π-Josephson junction film.

Prediction. $F_B/F_A = -1$ (sign reversal, same magnitude). Measurable with existing Casimir force apparatus [?].

Test 2: Critical Casimir enhancement at T_c

The critical Casimir effect (Fisher-de Gennes 1978, Hertlein et al. Nature 2008) amplifies vacuum forces near a continuous phase transition by $(\xi/d)^{3/2}$, where $\xi = \xi_0/\sqrt{1 - T/T_c}$ is the divergent correlation length.

For the π-junction (Nb, $T_c = 9.3\,\text{K}$, $\xi_0 = 38\,\text{nm}$, CuNi barrier $d = 2\,\text{nm}$): at $T/T_c = 0.9999$, $\xi \approx 4\,\mu\text{m}$ and $\xi/d \approx 2000$, giving $\sim 10^5\times$ Casimir enhancement. This makes the sign-reversal signal in Test 1 dramatically easier to detect: δf amplified by 10^5 relative to far from T_c.

Protocol. Sweep temperature through T_c and measure $\delta f(T)$. The sign of δf should *flip* at T_c (normal $\to$ π-phase), and the magnitude should diverge as $(1 - T/T_c)^{-3/4}$.

Direct gravitational detection remains challenging with a plain Nb junction ($F \sim 10^{-21}\,\text{N}$ for $10\,\text{cm}^2$). However, combining the critical enhancement with *magic-angle twisted bilayer graphene* (MATBG) offers a path.

MATBG at the magic angle ($\theta \approx 1.1°$) has flat electronic bands with divergent density of states, which amplify the Casimir response by $(v_F/v_F^*)^3 \sim 10^3$ (velocity renormalization from $v_F \to v_F^* \approx v_F/10$). MATBG also possesses an intrinsic π-Berry phase at the K-point.

Combined enhancement (MATBG moiré $\times$ QCP): at $T/T_c = 0.9999$ with moiré period $\sim 13\,\text{nm}$ and hBN spacer $d \approx 1.5\,\text{nm}$: $\Delta M/M \sim 10^{-4}$. The absolute force for $1\,\text{cm}^2$ is $\sim 10^{-19}\,\text{N}$ (mean-field QCP).

If the MATBG correlated-insulator transition is Berezinskii-Kosterlitz-Thouless (BKT) type (partial experimental evidence exists [?]), ξ diverges *exponentially*: at $\Delta T = 0.1\,\text{K}$, the required area drops to $\sim 2\,\text{cm}^2$—within reach of next-generation wafer-scale MATBG fabrication.

This provides a potential route to Phase 3 (direct gravitational detection) *without* Euler product amplification and *without* the associated safety concerns.

Test 3: The 182 vs. 1/3 discriminator

The Euler product test validates the spectral action framework via the Casimir energy ratio in a 2-prime-sieved BAW cavity. Both predictions are mathematically exact:

$$\text{Zeta:} \quad \frac{|\zeta_{\neg\{2,3\}}(-3)|}{|\zeta(-3)|} = |(1-2^3)(1-3^3)| = 182, \tag{7}$$

$$\text{Cutoff:} \quad \frac{\sum_{\gcd(n,6)=1}^{N} n^3}{\sum_{n=1}^{N} n^3} \to \left(1-\tfrac{1}{2}\right)\left(1-\tfrac{1}{3}\right) = \tfrac{1}{3}. \tag{8}$$

The 546-fold discrepancy arises because zeta regularization evaluates the Euler factor $(1 - p^{-s})$ at $s = -3$ (giving $1 - p^3$), while the cutoff evaluates it at $s = 0$ (giving

$1-1/p$). The former "sees" the arithmetic structure; the latter does not.

Standard Casimir experiments confirm zeta regularization for absolute energies, but *ratios between different spectral geometries* are untested. This experiment—at a cost of \$2,200—is the first such test. If 182: proceed to π-junction tests. If 1/3: the spectral action framework is falsified for non-standard mode spectra.

RELATION TO KNOWN PHYSICS

Why this is not trivially ruled out

One might object: a π-phase on cavity modes cannot affect gravity, because gravity couples to the stress-energy tensor $T_{\mu\nu}$, not to wave function phases.

In classical GR, this is correct. However, the spectral action principle *derives* $T_{\mu\nu}$ from the spectrum of D. If the spectrum includes a global π-phase, $T_{\mu\nu}$ itself is modified.

This is the distinction between:

- Standard GR: $T_{\mu\nu}$ is given, gravity follows.
- Spectral gravity: D is given, both $T_{\mu\nu}$ and $g_{\mu\nu}$ follow.

In the spectral framework, modifying D modifies everything simultaneously—matter *and* gravity.

Connection to dark energy

In previous work [**?**], we showed $\Omega_\Lambda = (d/\mathrm{cd})\,|\xi_{\neg 2}(-1)| = 2\pi/9$, where the sign of dark energy ($\zeta_{\neg 2}(-1) = +1/12 > 0$) follows from $p = 2$ muting of the fermionic vacuum ($\zeta(-1) = -1/12 < 0$).

The π-Berry phase result extends this: *the full fermionic vacuum $-\zeta(t)$ is gravitationally repulsive at all scales*, not just at the cosmological scale. Dark energy ($\Omega_\Lambda > 0$) is the cosmological manifestation of the same sign reversal.

DISCUSSION

Assumptions. The result depends on Connes' spectral action principle applied to the Bost-Connes spectral triple. Both are mathematically rigorous but physically unverified for real gravity. The proposed experiments test this directly.

If confirmed: π-Berry phase materials would constitute the first known "anti-gravity matter"—substances whose topological properties reverse the gravitational coupling. Applications include shielding, propulsion, and the inner structure of warp bubble architectures [**?**].

If refuted: the spectral action framework does not describe physical gravity at the level of vacuum mode phases, and the Bost-Connes identification $K(t) = \zeta(t)$ is mathematical rather than physical.

Either outcome would be a significant result for the foundations of quantum gravity.

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